Series & Binomial Theorem

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational] • Target: Arithmetic Progressions (AP)**

A particle moves along a straight line. Its velocity, v m/s, at time t seconds is given by $v = 3t - 2$. Find the displacement of the particle during the first 4 seconds of its motion.

The displacement of the particle can be found by integrating the velocity function with respect to time. The displacement s is given by: $s = \int v\,dt$ In this case, $v = 3t - 2$. $s = \int (3t - 2)\,dt$ Integrating, we get: $s = \frac{3}{2}t^2 - 2t + C$ To find the displacement during the first 4 seconds, we evaluate the definite integral from $t = 0$ to $t = 4$. We can assume $C = 0$ for calculating the change in displacement from $t = 0$. $s(4) - s(0) = \left[\frac{3}{2}t^2 - 2t\right]_0^4$.

2 **[Foundational] • Target: Geometric Progressions (GP)**

The diagram shows the graph of the function $y = f(x)$. The graph is transformed by a sequence of operations. The resulting graph, $y = g(x)$, is also shown. The function $f(x)$ is given by $f(x) = ax^2 + bx + c$, where $a, b,$ and c are constants. The vertex of the graph of $y = f(x)$ is at $(2, -1)$, and it passes through the point $(4, 3)$. The graph of $y = g(x)$ is obtained by translating the graph of $y = f(x)$ by $(-1\ 3)$ followed by a reflection in the y-axis. Find the values of $a, b,$ and c.

The vertex of the parabola $y = ax^2 + bx + c$ is at $\left(-\frac{b}{2a}, f\left(-\frac{b}{2a}\right)\right)$. We are given that the vertex is at $(2, -1)$. Therefore, $-\frac{b}{2a} = 2$, which implies $b = -4a$. The vertex form of the quadratic equation is $y = a(x - h)^2 + k$, where (h, k) is the vertex. So, $y = a(x - 2)^2 - 1$. The graph passes through $(4, 3)$. Substituting this point into the equation: $3 = a(4 - 2)^2 - 1 \Rightarrow 3 = a(2)^2 - 1 \Rightarrow 3 = 4a - 1 \Rightarrow 4a = 4 \Rightarrow a = 1$. Si.

3 **[Foundational] • Target: Sum to Infinity of a Convergent GP**

A committee of 3 people is to be selected from a group of 5 people. The number of ways to form this committee is given by the combination formula ${}^nC_r = \frac{n!}{r!\,(n-r)!}$. How many different committees can be formed?

The problem requires us to find the number of combinations of selecting 3 people from a group of 5. This is a direct application of the combination formula ${}^nC_r$, where n is the total number of items to choose from, and r is the number of items to choose. In this case, $n = 5$ and $r = 3$. Using the formula: ${}^5C_3 = \frac{5!}{3!\,(5-3)!} = \frac{5!}{3!\,2!} = \frac{5 \times 4 \times 3!}{3! \times (2 \times 1)} = \frac{5 \times 4}{2} = \frac{20}{2} = 10$. Therefore, there are 10 different committees that can be forme.

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4 [Foundational] • Target: The Binomial Theorem for Positive Integers

The diagram shows a right-angled triangle ABC, with the right angle at D. The altitude from D to the hypotenuse BC is drawn. Given that BD = 4 cm and DC = 9 cm, find the length of the altitude AD.

In a right-angled triangle, the altitude to the hypotenuse creates three similar triangles. Specifically, triangle ABD is similar to triangle CAD, and both are similar to the original triangle ABC. By the geometric mean theorem (or by using similarity ratios), the square of the altitude to the hypotenuse is equal to the product of the two segments it divides the hypotenuse into. Therefore, $AD^2 = BD \times DC$. Given $BD = 4$ cm and $DC = 9$ cm. $AD^2 = 4 \times 9$ $AD^2 = 36$ $AD = \sqrt{36}$ $AD = 6$ cm.

5 [Foundational] • Target: Arithmetic Progressions (AP)

A particle of mass 3 kg is at rest on a smooth horizontal surface. A horizontal force of magnitude P N is applied to the particle, causing it to accelerate at a rate of 4 m/s^2. Find the value of P.

According to Newton's Second Law of Motion, the net force acting on an object is equal to the product of its mass and acceleration, i.e., F = ma. In this case, the mass of the particle is m = 3 kg and its acceleration is a = 4 m/s^2. The applied force P is the net force acting on the particle, as the surface is smooth and horizontal, meaning there are no other horizontal forces to consider. Therefore, P = ma = (3 kg) * (4 m/s^2) = 12 N. Thus, the value of P is 12 N.

6 [Foundational] • Target: Geometric Progressions (GP)

The two-way table shows the results of a survey of 100 students on their preferred mode of transport to school and their year group. | | Car | Bus | Walk | Total | |--------------|-----|-----|------|-------| | Year 1 | 15 | 20 | 10 | 45 | | Year 2 | 25 | 18 | 12 | 55 | | Total | 40 | 38 | 22 | 100 | What is the probability that a randomly selected student is a Year 1 student who prefers to travel by car?

The total number of students surveyed is 100. The number of Year 1 students who prefer to travel by car is 15. Therefore, the probability of selecting a Year 1 student who prefers to travel by car is the number of Year 1 students who prefer to travel by car divided by the total number of students. P(Year 1 and Car) = Number of (Year 1 and Car) / Total Number of Students = 15 / 100 = 0.15.

7 [Foundational] • Target: Sum to Infinity of a Convergent GP

A 3 kg particle on a rough surface, $\mu = 0.4$. Horizontal force F N. (a) Find the limiting friction. (b) Acceleration when $F = 20$ N. (c) Time to travel 5 m from rest.

(a) $F_f = 0.4 \times 3 \times 9.8 = 11.76$ N. (b) $a = \frac{20 - 11.76}{3} \approx 2.75$ m s^{-2}. (c) $t = \sqrt{10/2.75} \approx 1.91$ s. Key Mathematical Principle & Verification: In exam revision, always verify that the final result satisfies all domain restrictions, boundary conditions, and the original governing equation by direct substitution into the prompt.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: The Binomial Theorem for Positive Integers**

A factory tests 30 independent components, each with a fixed probability of being defective; X is the number of defective components. Is X binomially distributed?

The trials are fixed in number, independent, and each has the same probability of success, so the conditions for a binomial distribution are satisfied. Key Mathematical Principle & Verification: In exam revision, always verify that the final result satisfies all domain restrictions, boundary conditions, and the original governing equation by direct substitution into the prompt.

9 **[Intermediate]** • **Target: Arithmetic Progressions (AP)**

A survey asks 50 randomly and independently selected people the same yes/no question; X is the number who say yes. Is X binomially distributed?

The trials are fixed in number, independent, and each has the same probability of success, so the conditions for a binomial distribution are satisfied. Key Mathematical Principle & Verification: In exam revision, always verify that the final result satisfies all domain restrictions, boundary conditions, and the original governing equation by direct substitution into the prompt.

10 **[Intermediate]** • **Target: Geometric Progressions (GP)**

A machine has a useful power output of 883 W and wastes 367 W as heat. Find its efficiency as a percentage, correct to 1 decimal place.

Total input $= 883 + 367 = 1250$ W. Efficiency $= \frac{\text{useful output}}{\text{input}} \times 100\% = \frac{883}{1250} \times 100\% \approx 70.6\%$ Key Mathematical Principle & Verification: In exam revision, always verify that the final result satisfies all domain restrictions, boundary conditions, and the original governing equation by direct substitution into the prompt.

11 **[Intermediate]** • **Target: Sum to Infinity of a Convergent GP**

Solve $\cos(x) = -\frac{\sqrt{2}}{2}$ for $0^\circ \le x \le 360^\circ$.

The reference angle gives one solution and CAST-rule symmetry gives the second: $x = 135^\circ$ or $x = 225^\circ$ within $[0^\circ, 360^\circ]$. Key Mathematical Principle & Verification: In exam revision, always verify that the final result satisfies all domain restrictions, boundary conditions, and the original governing equation by direct substitution into the prompt.

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12 **[Intermediate]** • **Target: The Binomial Theorem for Positive Integers**

Solve $\cos(x) = -\frac{1}{2}$ for $0^\circ \leq x \leq 360^\circ$.

The reference angle gives one solution and CAST-rule symmetry gives the second: $x = 120^\circ$ or $x = 240^\circ$ within $[0^\circ, 360^\circ]$. Key Mathematical Principle & Verification: In exam revision, always verify that the final result satisfies all domain restrictions, boundary conditions, and the original governing equation by direct substitution into the prompt.

13 **[Intermediate]** • **Target: Arithmetic Progressions (AP)**

Solve $\tan(x) = -\frac{\sqrt{3}}{3}$ for $0^\circ \leq x \leq 360^\circ$.

The reference angle gives one solution and CAST-rule symmetry gives the second: $x = 150^\circ$ or $x = 330^\circ$ within $[0^\circ, 360^\circ]$. Key Mathematical Principle & Verification: In exam revision, always verify that the final result satisfies all domain restrictions, boundary conditions, and the original governing equation by direct substitution into the prompt.

14 **[Intermediate]** • **Target: Geometric Progressions (GP)**

Solve $\tan(x) = \frac{\sqrt{3}}{3}$ for $0^\circ \leq x \leq 360^\circ$.

The reference angle gives one solution and CAST-rule symmetry gives the second: $x = 30^\circ$ or $x = 210^\circ$ within $[0^\circ, 360^\circ]$. Key Mathematical Principle & Verification: In exam revision, always verify that the final result satisfies all domain restrictions, boundary conditions, and the original governing equation by direct substitution into the prompt.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Sum to Infinity of a Convergent GP** • **Cross-Lesson: 7.01 Statistics & Probability**

A particle of mass 5 kg slides down a rough plane inclined at 30° to the horizontal, experiencing a constant friction force of 0 N. Taking $g = 9.8$ m/s^2, find its acceleration down the slope.

$W = mg = 5 \times 9.8 = 49.00$ N. Component along slope $= W\sin 30^\circ \approx 24.50$ N. Net force ≈ 24.50 N. $a = F/m \approx 4.90$ m/s^2 Key Mathematical Principle & Verification: In exam revision, always verify that the final result satisfies all domain restrictions, boundary conditions, and the original governing equation by direct substitution into the prompt.

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16 **[Challenging]** • **Target: The Binomial Theorem for Positive Integers** • **Cross-Lesson: 8.01 Functions: Composite & Inverse**

Evaluate $\int_1^2 (6x^2 - 5x + 1)\,dx$.

$\int (6x^2 - 5x + 1)dx = 2x^3 - 5/2x^2 + x$. Evaluating from 1 to 2: $[2x^3 - 5/2x^2 + x]_1^2 = 15/2$ Key Mathematical Principle & Verification: In exam revision, always verify that the final result satisfies all domain restrictions, boundary conditions, and the original governing equation by direct substitution into the prompt.

17 **[Challenging]** • **Target: Arithmetic Progressions (AP)** • **Cross-Lesson: 9.01 Circular Measure**

Evaluate $\int_{-2}^4 (3x^2 - 3x - 1)\,dx$.

$\int (3x^2 - 3x - 1)dx = x^3 - 3/2x^2 - x$. Evaluating from -2 to 4: $[x^3 - 3/2x^2 - x]_{-2}^4 = 48$ Key Mathematical Principle & Verification: In exam revision, always verify that the final result satisfies all domain restrictions, boundary conditions, and the original governing equation by direct substitution into the prompt.

18 **[Challenging]** • **Target: Geometric Progressions (GP)** • **Cross-Lesson: 1.01 Indices, Surds & Quadratics**

Given $X \sim N(\mu, \sigma^2)$, $P(X < 99) = 0.8$ and $P(X < 83) = 0.2$, find μ, correct to the nearest integer.

Using $z_1 = 0.8416$, $z_2 = -0.8416$ and solving $x_1 = \mu + z_1\sigma$, $x_2 = \mu + z_2\sigma$ simultaneously gives $\mu \approx 91$. Key Mathematical Principle & Verification: In exam revision, always verify that the final result satisfies all domain restrictions, boundary conditions, and the original governing equation by direct substitution into the prompt.

19 **[Challenging]** • **Target: Sum to Infinity of a Convergent GP** • **Cross-Lesson: 11.01 Kinematics**

Evaluate $\int_1^3 (6x^2 + 2x + 3)\,dx$.

$\int (6x^2 + 2x + 3)dx = 2x^3 + x^2 + 3x$. Evaluating from 1 to 3: $[2x^3 + x^2 + 3x]_1^3 = 66$ Key Mathematical Principle & Verification: In exam revision, always verify that the final result satisfies all domain restrictions, boundary conditions, and the original governing equation by direct substitution into the prompt.

20 **[Challenging]** • **Target: The Binomial Theorem for Positive Integers** • **Cross-Lesson: 12.01 Forces & Equilibrium**

A particle of mass m kg is placed on a rough inclined plane. The plane is inclined at an angle α to the horizontal, where $\tan\alpha = \frac{3}{4}$. The particle is on the point of sliding down the plane. The coefficient of static friction between the particle and the plane is μ_s. Given that the acceleration due to gravity is $g = 9.8$ m/s^2, find the value of μ_s.

When a particle is on the point of sliding down an inclined plane, the angle of inclination α is equal to the angle of friction ϕ, where $\tan\phi = \mu_s$. Given that $\tan\alpha = \frac{3}{4}$, and the particle is on the point of sliding, we have $\alpha = \phi$. Therefore, $\mu_s = \tan\phi = \tan\alpha = \frac{3}{4}$.

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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Arithmetic Progressions (AP)

An arithmetic progression has first term $a = 7$ and common difference $d = 4$. Find the 20th term u_{20}.

a. **83** ✓ **b.** 87

c. 79 **d.** 80

$u_n = a + (n-1)d \Rightarrow u_{20} = 7 + 19(4) = 7 + 76 = 83$.

22 [Foundational] • Target: Arithmetic Progressions (AP)

Find the sum of the first 25 terms of the AP: $3, 8, 13, 18, \ldots$

a. **1575** ✓ **b.** 1625

c. 1500 **d.** 1550

$a = 3, d = 5$. $S_{25} = \frac{25}{2}[2(3) + 24(5)] = \frac{25}{2}[6 + 120] = \frac{25}{2}(126) = 25 \times 63 = 1575$.

23 [Foundational] • Target: Arithmetic Progressions (AP)

In an AP, the 5th term is 18 and the 12th term is 39. Find the first term a and common difference d.

a. $a = 6, d = 3$ ✓ **b.** $a = 3, d = 3$

c. $a = 9, d = 3$ **d.** $a = 6, d = 4$

$a + 4d = 18$ and $a + 11d = 39$. Subtracting: $7d = 21 \Rightarrow d = 3$. Then $a = 18 - 4(3) = 6$.

24 [Foundational] • Target: Geometric Progressions (GP)

A geometric progression has first term $a = 5$ and common ratio $r = 2$. Find the 8th term u_8.

a. **640** ✓ **b.** 1280

c. 320 **d.** 512

$u_n = ar^{n-1} \Rightarrow u_8 = 5(2)^7 = 5(128) = 640$.

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25 [Foundational] • Target: Geometric Progressions (GP)

Find the sum of the first 6 terms of the GP: $3, -6, 12, -24, \ldots$

a. -63 ✓

b. 63

c. -127

d. -65

$a = 3, r = -2$. $S_6 = \frac{3(1-(-2)^6)}{1-(-2)} = \frac{3(1-64)}{3} = 1 - 64 = -63$.

26 [Foundational] • Target: Geometric Progressions (GP)

The 2nd term of a GP is 12 and the 5th term is 324. Find the common ratio r.

a. 3 ✓

b. 2

c. 4

d. $\sqrt{3}$

$\frac{u_5}{u_2} = \frac{ar^4}{ar} = r^3 = \frac{324}{12} = 27 \Rightarrow r = 3$.

27 [Foundational] • Target: Sum to Infinity of a Convergent GP

State the condition for the sum to infinity of a geometric progression to exist.

a. $|r| < 1$ ✓

b. $r > 1$

c. $r \leq 1$

d. $|r| > 1$

The infinite geometric series converges if and only if the absolute value of the common ratio is strictly less than 1: $|r| < 1$.

28 [Foundational] • Target: Sum to Infinity of a Convergent GP

Find the sum to infinity of the GP: $16, 12, 9, 6.75, \ldots$

a. **64** ✓

b. 48

c. 32

d. 72

$a = 16, r = 12/16 = 3/4$. Since $|3/4| < 1$, $S_\infty = \frac{a}{1-r} = \frac{16}{1-3/4} = \frac{16}{1/4} = 64$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Sum to Infinity of a Convergent GP

Express the recurring decimal $0.\overline{27} = 0.272727\ldots$ as an exact fraction in lowest terms using an infinite geometric series.

a. $\frac{3}{11}$ ✓

b. $\frac{27}{100}$

c. $\frac{9}{33}$

d. $\frac{7}{25}$

$0.27 + 0.0027 + \ldots$ has $a = 0.27 = 27/100$ and $r = 1/100$. $S_\infty = \frac{27/100}{1 - 1/100} = \frac{27/100}{99/100} = \frac{27}{99} = \frac{3}{11}$.

30 [Intermediate] • Target: The Binomial Theorem for Positive Integers

Find the coefficient of x^3 in the expansion of $(2 + 3x)^5$.

a. 1080 ✓

b. 720

c. 540

d. 1440

General term is $\binom{5}{r}2^{5-r}(3x)^r$. For x^3, $r = 3$: $\binom{5}{3}2^2(3)^3 = 10 \times 4 \times 27 = 1080$.

31 [Intermediate] • Target: The Binomial Theorem for Positive Integers

Find the term independent of x in the expansion of $\left(x^2 + \frac{2}{x}\right)^6$.

a. **240** ✓

b. 60

c. 160

d. 120

General term is $\binom{6}{r}(x^2)^{6-r}\left(\frac{2}{x}\right)^r = \binom{6}{r}2^r x^{12-2r-r} = \binom{6}{r}2^r x^{12-3r}$. Independent when $12 - 3r = 0 \Rightarrow r = 4$. Term is $\binom{6}{4}2^4 = 15 \times 16 = 240$.

32 [Intermediate] • Target: The Binomial Theorem for Positive Integers

In the expansion of $(1 + ax)^6$, the coefficient of x^2 is 60. Given $a > 0$, find a.

a. **2** ✓

b. 4

c. 3

d. 5

Coefficient of x^2 is $\binom{6}{2}a^2 = 15a^2$. $15a^2 = 60 \Rightarrow a^2 = 4 \Rightarrow a = 2$ (since $a > 0$).

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33 [Intermediate] • Target: Arithmetic Progressions (AP)

A geometric series has first term $a = -10$ and common ratio $r = 1/2$. Find S_6, the sum of the first 6 terms.

a. **-315/16 ✓**

b. -155/8

c. -475/16

d. -635/32

$S_6 = \frac{a(1-r^n)}{1-r} = \frac{-10(1-(1/2)^6)}{1-(1/2)} = -315/16$

34 [Intermediate] • Target: Geometric Progressions (GP)

A geometric series has first term $a = 13$ and common ratio $r = 5/2$. Find S_5, the sum of the first 5 terms.

a. 67431/32

b. **13403/16 ✓**

c. 2639/8

d. 13611/16

$S_5 = \frac{a(1-r^n)}{1-r} = \frac{13(1-(5/2)^5)}{1-(5/2)} = 13403/16$

35 [Intermediate] • Target: Sum to Infinity of a Convergent GP

A geometric series has first term $a = -3$ and common ratio $r = 2/3$. Find S_5, the sum of the first 5 terms.

a. **-211/27 ✓**

b. -292/27

c. -65/9

d. -665/81

$S_5 = \frac{a(1-r^n)}{1-r} = \frac{-3(1-(2/3)^5)}{1-(2/3)} = -211/27$

36 [Intermediate] • Target: The Binomial Theorem for Positive Integers

A geometric series has first term $a = 6$ and common ratio $r = -4/3$. Find S_5, the sum of the first 5 terms.

a. -50/9

b. -962/81

c. 524/27

d. **362/27 ✓**

$S_5 = \frac{a(1-r^n)}{1-r} = \frac{6(1-(-4/3)^5)}{1-(-4/3)} = 362/27$

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37 [Intermediate] • Target: Arithmetic Progressions (AP)

Find the sum of the first 40 terms of an arithmetic sequence with first term -11 and common difference -2.

a. -2000 ✓

b. -2004

c. -1994

d. -2003

$S_{40} = \frac{n}{2}(2a + (n-1)d) = \frac{40}{2}(2(-11) + (40-1)(-2)) = -2000$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Geometric Progressions (GP) • Cross-Lesson: 1.01 Indices, Surds & Quadratics

A geometric series has first term $a = 6$ and common ratio $r = 3/2$. Find S_6, the sum of the first 6 terms.

a. 1995/16 ✓

b. 2091/16

c. 633/8

d. 6177/32

$S_6 = \frac{a(1-r^n)}{1-r} = \frac{6(1-(3/2)^6)}{1-(3/2)} = 1995/16$

39 [Challenging] • Target: Sum to Infinity of a Convergent GP • Cross-Lesson: 11.01 Kinematics

A geometric series has first term $a = 2$ and common ratio $r = 5/2$. Find S_6, the sum of the first 6 terms.

a. 5187/16 ✓

b. 5219/16

c. 25999/32

d. 1031/8

$S_6 = \frac{a(1-r^n)}{1-r} = \frac{2(1-(5/2)^6)}{1-(5/2)} = 5187/16$

40 [Challenging] • Target: The Binomial Theorem for Positive Integers • Cross-Lesson: 12.01 Forces & Equilibrium

A geometric series has first term $a = 1$ and common ratio $r = 4/3$. Find S_6, the sum of the first 6 terms.

a. 3367/243 ✓

b. 14197/729

c. 3610/243

d. 781/81

$S_6 = \frac{a(1-r^n)}{1-r} = \frac{(1-(4/3)^6)}{1-(4/3)} = 3367/243$

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41 [Challenging] • Target: Arithmetic Progressions (AP) • Cross-Lesson: 13.01 Newton's Laws

Find the sum of the first 40 terms of an arithmetic sequence with first term -9 and common difference -2.

a. -1920 ✓

b. -1926

c. -1916

d. -1923

$S_{40} = \frac{n}{2}(2a + (n-1)d) = \frac{40}{2}(2(-9) + (40-1)(-2)) = -1920$

42 [Challenging] • Target: Geometric Progressions (GP) • Cross-Lesson: 14.01 Energy, Work & Power

Find the sum of the first 46 terms of an arithmetic sequence with first term -9 and common difference -7.

a. -7662

b. -7659 ✓

c. -7668

d. -7669

$S_{46} = \frac{n}{2}(2a + (n-1)d) = \frac{46}{2}(2(-9) + (46-1)(-7)) = -7659$

43 [Challenging] • Target: Sum to Infinity of a Convergent GP • Cross-Lesson: 15.01 Representation of Data

Find the sum of the first 17 terms of an arithmetic sequence with first term -12 and common difference -3.

a. -612 ✓

b. -608

c. -607

d. -613

$S_{17} = \frac{n}{2}(2a + (n-1)d) = \frac{17}{2}(2(-12) + (17-1)(-3)) = -612$

44 [Challenging] • Target: The Binomial Theorem for Positive Integers • Cross-Lesson: 16.01 Permutations & Combinations

A geometric series has first term $a = 3$ and common ratio $r = 1/2$. Find S_5, the sum of the first 5 terms.

a. 93/16 ✓

b. 141/16

c. 45/8

d. 189/32

$S_5 = \frac{a(1-r^n)}{1-r} = \frac{3(1-(1/2)^5)}{1-(1/2)} = 93/16$

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45 [Challenging] • Target: Arithmetic Progressions (AP) • Cross-Lesson: 17.01 Discrete Random Variables

Find the sum of the first 32 terms of an arithmetic sequence with first term -12 and common difference 7.

a. 3074

b. 3077

c. **3088** ✓

d. 3078

$S_{32} = \frac{n}{2}(2a + (n-1)d) = \frac{32}{2}(2(-12) + (32-1)(7)) = 3088$

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